Methods: Kinetic Segregation Model of TCR–CD45 Spatial Exclusion on a 2D Membrane Patch

Based on Neve-Oz, Sherman & Raveh, *Frontiers in Immunology*, 2024

1 Model overview

The kinetic segregation (KS) model simulates the spatial segregation of T-cell receptor (TCR) and CD45 molecules on a 2 μm $\times$ 2 μm membrane patch. A 2D membrane height field $h(x, y)$ is discretised on a regular grid of spacing $\Delta x = L/N_{\text{grid}}$, where $L = 2000$ nm and N_{grid} is typically 64. Static pMHC ligands (N_{pMHC}, default 200) are placed on the opposing antigen-presenting cell (APC) surface and gate TCR–pMHC binding: TCR molecules (N_{TCR}, default 50) are attracted to tight-contact regions (low h) *only in grid cells that contain at least one pMHC*. CD45 molecules (N_{CD45}, default 100) are sterically excluded from regions where h is smaller than the CD45 ectodomain height $h_{\text{CD45}} = 50$ nm.

The system is evolved with a Metropolis–Hastings Monte Carlo (MC) algorithm at thermal energy $k_B T = 1$ (all energies are expressed in units of $k_B T$).

2 System energy

The total energy of the system is the sum of three contributions:

$$E_{\text{total}} = E_{\text{bend}}\big[h\big] + \sum_{i=1}^{N_{\text{TCR}}} \mathbb{1}_{\text{pMHC}}(\mathbf{r}_i)\, E_{\text{TCR}}\big(h(\mathbf{r}_i)\big) + \sum_{j=1}^{N_{\text{CD45}}} E_{\text{CD45}}\big(h(\mathbf{r}_j)\big) \tag{1}$$

where $\mathbf{r}_i$ and $\mathbf{r}_j$ are the lateral positions of individual TCR and CD45 molecules, respectively, $h(\mathbf{r})$ is the membrane height at that position (obtained by nearest-grid-point interpolation), and $\mathbb{1}_{\text{pMHC}}(\mathbf{r})$ is unity when the grid cell at $\mathbf{r}$ contains at least one pMHC and zero otherwise.

3 Potential energy terms

Table 1 summarises all energy contributions, their functional forms, and default parameter values.

3.1 Membrane bending energy

The bending energy follows the Helfrich Hamiltonian discretised on a square grid with periodic boundary conditions. The discrete Laplacian at grid point (i, j) is computed via the standard five-point stencil:

$$\nabla_d^2\, h_{ij} = \frac{h_{i-1,j} + h_{i+1,j} + h_{i,j-1} + h_{i,j+1} - 4\, h_{ij}}{\Delta x^2} \tag{2}$$

with indices taken modulo N_{grid} (periodic boundaries). The full bending energy is then:

$$E_{\text{bend}} = \frac{\kappa}{2} \sum_{i=0}^{N_{\text{grid}}-1} \sum_{j=0}^{N_{\text{grid}}-1} \big(\nabla_d^2\, h_{ij}\big)^2\, \Delta x^2 \tag{3}$$

Table 1: Energy terms in the kinetic segregation model. All energies are in units of k_BT; lengths are in nm.

Term	Expression	Parameters	Description
Membrane bending	$E_{\text{bend}} = \frac{\kappa}{2}\sum_{i,j}\left(\nabla_d^2 h_{ij}\right)^2 \Delta x^2$	κ: bending rigidity (k_BT); $\Delta x = L/N_{\text{grid}}$	Helfrich bending energy on a discrete grid with periodic BCs
TCR–pMHC binding	$E_{\text{TCR}}(h) = -U_{\text{assoc}} \exp\Big(-\frac{(h-h_0^{\text{TCR}})^2}{2\sigma_{\text{bind}}^2}\Big)$	$U_{\text{assoc}} = 20\ k_BT$; $\sigma_{\text{bind}} = 3$ nm	Gaussian attractive well centred at the bond length h_0^{TCR}
CD45 steric repulsion	$E_{\text{CD45}}(h) = \begin{cases} \frac{1}{2}\, k_{\text{rep}}\,(h_{\text{CD45}}-h)^2 & h < h_{\text{CD45}} \\ 0 & h \geq h_{\text{CD45}} \end{cases}$	$k_{\text{rep}} = 1.0\ k_BT/\text{nm}^2$; $h_{\text{CD45}} = 50$ nm	Harmonic repulsive wall; excludes CD45 from tight contact

3.2 TCR–pMHC binding potential

TCR molecules are attracted to regions of low membrane height (tight contact between the T-cell and antigen-presenting cell), modelling TCR–pMHC engagement. The potential is a Gaussian well centred at the resting TCR–pMHC bond length $h_0^{\text{TCR}} = 13$ nm—*not* at $h = 0$. A bound TCR is equally disfavoured when the membranes are too far apart and when they are squeezed closer than the bond allows:

$$E_{\text{TCR}}(h) \;=\; -U_{\text{assoc}} \exp\left(-\frac{(h-h_0^{\text{TCR}})^2}{2\,\sigma_{\text{bind}}^2}\right) \tag{4}$$

where $U_{\text{assoc}} = 20\ k_BT$ is the association energy depth and $\sigma_{\text{bind}} = 3$ nm is the spatial width of the binding well.

3.3 CD45 steric repulsion

CD45 has a large extracellular domain ($\sim$35 nm) and is sterically excluded from tight-contact zones. This is modelled as a one-sided harmonic potential:

$$E_{\text{CD45}}(h) \;=\; \begin{cases} \dfrac{1}{2}\, k_{\text{rep}} \left(h_{\text{CD45}} - h\right)^2 & \text{if } h < h_{\text{CD45}} \\ 0 & \text{otherwise} \end{cases} \tag{5}$$

with $k_{\text{rep}} = 1.0\ k_BT/\text{nm}^2$ and $h_{\text{CD45}} = 50$ nm.

4 Monte Carlo algorithm

The simulation uses a Metropolis–Hastings MC scheme with two alternating update phases per sweep. Each *sweep* updates every degree of freedom once (all molecules and all grid cells), ensuring ergodic sampling of the configuration space.

Algorithm 1 gives the complete pseudocode.

4.1 Step sizes

Step sizes are derived from Brownian dynamics. A stability-limited time step Δt is computed as

$$\Delta t = \frac{\Delta x^2}{2\, D_h\, \kappa}\, \alpha_{\text{safety}} \tag{6}$$

with $\alpha_{\text{safety}} = 0.05$ (`MAX_BENDING_DE_PER_STEP`), the maximum typical bending ΔE per grid point per step. A second reduction then applies: $\Delta t \leftarrow \min(\Delta t,\ \ell^2/(8D_{\text{mol}}))$, where ℓ is the smallest resolved length among Δx, σ_r (gaussian binding with pMHC present) and the molecular repulsion cutoff. This keeps a proposed lateral move small relative to every feature it must resolve. The molecular and membrane height step sizes are then

$$\sigma_{\text{step}} = \sqrt{2\,D_{\text{mol}}\,\Delta t}, \qquad \sigma_h = \sqrt{2\,D_h\,\Delta t} \tag{7}$$

where $D_{\text{mol}} = 10^4\ \text{nm}^2/\text{s}$ and $D_h = 5 \times 10^4\ \text{nm}^2/\text{s}$ are the molecular and membrane height diffusion coefficients. An alternative "paper" mode uses fixed $\Delta t = 0.01$ s and $\sigma_h = 1$ nm.

4.2 Efficient bending energy update

Rather than recomputing the full bending energy (an $O(N_{\text{grid}}^2)$ operation), the change in bending energy when a single grid cell (i,j) is updated is computed in $O(1)$ time. Only the five Laplacian stencil cells affected by the change—(i,j) itself and its four periodic neighbours—need to be considered:

$$\Delta E_{\text{bend}} = \frac{\kappa}{2}\,\Delta x^2 \sum_{(a,b)\in\mathcal{N}(i,j)} \left[\left(\nabla_d^2 h_{ab}^{\text{new}}\right)^2 - \left(\nabla_d^2 h_{ab}^{\text{old}}\right)^2\right] \tag{8}$$

where $\mathcal{N}(i,j) = \{(i,j),\ (i \pm 1, j),\ (i, j \pm 1)\}$ with periodic wrapping.

4.3 Checkerboard decomposition for Phase 2

Phase 2 (membrane height updates) uses a red–black checkerboard decomposition. Grid cells are coloured so that each cell of a given colour has only opposite-colour neighbours in the five-point stencil. All same-colour cells can then be updated simultaneously without data races, because their energy deltas depend only on the frozen opposite-colour neighbours.

Each sweep processes both colours sequentially (red then black), ensuring that every grid cell is updated exactly once. On the CPU this proceeds serially within each colour; on the GPU, all cells of a given colour are dispatched in parallel.

4.4 Initial conditions

- The membrane height field is initialised uniformly at $h = h_{\text{init}} = 70$ nm, with a central disc (radius $\sim N_{\text{grid}}/8$ cells) depressed to $h_0^{\text{TCR}} = 13$ nm to seed a tight-contact zone.
- pMHC ligands are placed on the APC surface—either uniformly across the patch, or within a centred disc of radius $L/3$ ("inner-circle" mode). Uniform placement is the default; note that inner-circle is required for the centre-referenced depletion metrics to be meaningful. They remain static throughout the simulation.
- TCR molecules are co-located with randomly chosen pMHC positions, with a small Gaussian jitter (σ_{bind}), so that each TCR starts near a cognate pMHC. When no pMHC are present, TCRs are placed with a Gaussian bias towards the patch centre (spread $L/6$).
- CD45 molecules are placed uniformly at random across the patch.

4.5 Output

The primary observable is the *depletion zone width*, defined as the difference between the median radial distance from the patch centre for CD45 molecules and that for TCR molecules:

$$w_{\text{depletion}} = \max\Big(0,\ \widetilde{r}_{\text{CD45}} - \widetilde{r}_{\text{TCR}}\Big) \tag{9}$$

where $\tilde{r}$ denotes the median over all molecules of that species. The use of the median rather than a percentile-based measure provides robustness at small molecule counts.

5 Simulation parameters

Table 2 lists all tunable parameters and their default values from Supplementary Table S1 of the reference.

Table 2: Default simulation parameters.

Parameter	Symbol	Default value
Patch size	L	2000 nm (2 μm)
Grid resolution	N_{grid}	64
Grid spacing	Δx	$L/N_{\text{grid}} \approx 31.25$ nm (but see §5.1)
Number of TCR	N_{TCR}	125
Number of CD45	N_{CD45}	500
Number of pMHC	N_{pMHC}	auto: 300/μm^2 over the placement area
Association energy	U_{assoc}	20 k_BT
Binding well width	σ_{bind}	3 nm
CD45 ectodomain	h_{CD45}	50 nm
Repulsive constant	k_{rep}	1.0 k_BT/nm^2
Initial height	h_{init}	70 nm
TCR contact height	h_0^{TCR}	13 nm
Mol. diffusion	D_{mol}	10^4 nm^2/s
Height diffusion	D_h	5×10^4 nm^2/s
Safety factor	α_{safety}	0.05
Bending rigidity	κ	Swept (input parameter, k_BT)
Simulation time	t	Swept (input parameter)
Random seed	—	42 (per-point FNV-1a hash derived)

5.1 Grid resolution and the pMHC influence field

In `gaussian` binding mode the TCR well at each cell is scaled by a pMHC influence weight $W = \min\bigl(1, \sum_k \exp(-r_k^2/2\sigma_r^2)\bigr)$, summed over pMHC within $3\sigma_r$ of the cell centre, with $\sigma_r = 2$ nm by default.

This makes Δx a physical parameter rather than a discretisation choice. When $\Delta x \gg \sigma_r$ no cell centre lies within the cutoff, W is identically zero, and the attractive term vanishes—so a seeded contact simply relaxes to h_{init}. Measured at fixed physical time, the mean height inside the contact disc ends at 63.6 nm for $\Delta x = 41.7$ nm versus 20.2 nm for $\Delta x = 7.8$ nm. Configurations intended to exercise binding should therefore keep Δx within roughly a factor of a few of σ_r; the behaviour sweep in `experiments/ks_behavior_sweep/` uses $L = 500$ nm with $N_{\text{grid}} = 100$ ($\Delta x = 5$ nm) for this reason.

6 Implementation

The model is implemented as a portable C99 simulation core with a C++20 command-line interface and an optional GPU backend. A single binary (`ks_gpu`) provides both CPU and GPU execution modes. The build system uses CMake with platform detection: Apple Metal on macOS, with a planned CUDA backend for Linux.

6.1 Code architecture

The source is organised into four layers (Table 3).

Table 3: Source code organisation.

Layer	Language	Files / purpose
Simulation core	C99	`simulation.c`, `potentials.c`, `rng.c` — MC sweep, molecular potentials, PCG64 RNG. Fully portable; no platform dependencies.
Shared physics	C99 / MSL	`ks_physics.h` — shared `static inline` float functions for TCR binding, CD45 repulsion, Laplacian, and bending-energy delta. Preprocessor guards (`__METAL_VERSION__`, `__CUDA_ARCH__`) select Metal address-space qualifiers or CUDA decorators.
GPU backend	Objective-C / MSL	`gpu_engine.h` (backend-neutral C API), `metal_engine.m` (Metal implementation), `shaders.metal` (GPU compute kernels).
CLI	C++20	`main.cpp` — argument parsing, JSON I/O (nlohmann/json, header-only), directory management (`std::filesystem`), deterministic seed derivation (FNV-1a hash). No platform-specific dependencies.

The shared physics header (`ks_physics.h`) is the single source of truth for the four float-precision energy functions used by both the CPU Phase 2 path (`simulation.c`) and the GPU evaluate kernel (`shaders.metal`). This eliminates the previous code duplication and guarantees CPU–GPU equivalence by construction.

6.2 GPU backend interface

The GPU backend is accessed through a four-function C API defined in `gpu_engine.h`: `gpu_engine_create`, `gpu_engine_destroy`, `gpu_engine_h_ptr`, and `gpu_engine_grid_update`. The interface uses an opaque `void*` handle, allowing different backends (Metal, CUDA) to be linked at compile time without changing the simulation core.

6.3 CPU mode

In CPU mode, Phase 1 (molecular moves) and Phase 2 (grid height updates) both execute sequentially. Random numbers are generated with the PCG64 algorithm, seeded deterministically from the user-provided seed. Phase 2 uses the checkerboard decomposition described above but updates each colour's cells serially.

6.4 GPU acceleration (Metal)

On Apple Silicon, Phase 2 is offloaded to the GPU via Metal compute shaders. The pipeline consists of four kernel dispatches per colour:

1. **Propose**: Each thread proposes a new height for one cell using a Philox counter-based RNG.
2. **Snapshot**: Copies the full height field to a frozen buffer so that the evaluate kernel reads consistent neighbour values.

3. **Evaluate**: Computes the energy delta (bending + molecular contributions) and the Metropolis acceptance criterion.

4. **Apply**: Writes accepted proposals back to the authoritative height buffer.

Phase 1 remains on the CPU (few molecules, inherently sequential). The height field resides in a Metal shared buffer (`MTLResourceStorageModeShared`), allowing zero-copy access from both CPU and GPU on unified memory.

A grid-substep parameter K allows batching K Phase 2 sweeps per Phase 1 molecular move, reducing CPU–GPU synchronisation overhead. All $K \times 8$ kernel dispatches are encoded into a single Metal command buffer before committing.

6.5 Performance

Table 4 shows representative timings on Apple M3 for 500-step simulations.

Table 4: CPU vs GPU wall-clock time (500 steps, $\kappa = 20$, $N_{\text{TCR}} = 125$, $N_{\text{CD45}} = 500$).

N_{grid}	CPU (s)	GPU (s)	Speedup
64	0.14	0.06	2.3×
128	0.50	0.06	8.7×
256	1.78	0.07	25×
512	6.29	0.09	69×

References

Y. Neve-Oz, E. Sherman, and B. Raveh. Bayesian metamodeling of early T-cell antigen receptor signaling accounts for its nanoscale activation patterns. *Frontiers in Immunology*, 2024.

Algorithm 1 Metropolis Monte Carlo for kinetic segregation

Require: N_{steps}, membrane grid h, positions $\{\mathbf{r}_i^{\text{TCR}}\}$, $\{\mathbf{r}_j^{\text{CD45}}\}$, parameters $\kappa, U_{\text{assoc}}, \sigma_{\text{bind}}, h_{\text{CD45}}, k_{\text{rep}}$

Ensure: Final configuration $\{h, \mathbf{r}^{\text{TCR}}, \mathbf{r}^{\text{CD45}}\}$

for $s = 1$ **to** N_{steps} **do**

▷ **Phase 1: Molecular moves**

  for each molecule k in TCR $\cup$ CD45 **do**

    $\mathbf{r}_{\text{old}} \leftarrow \mathbf{r}_k$

    $E_{\text{old}} \leftarrow E_{\text{mol}}(\mathbf{r}_k, h)$ ▷ Eq. 4 or 5

    $\mathbf{r}_k \leftarrow \mathbf{r}_k + \mathcal{N}(0,\, \sigma_{\text{step}}^2)\ \hat{\mathbf{e}}$ ▷ Gaussian displacement in x, y

    Wrap $\mathbf{r}_k$ with periodic boundaries (mod L)

    $E_{\text{new}} \leftarrow E_{\text{mol}}(\mathbf{r}_k, h)$

    $\Delta E \leftarrow E_{\text{new}} - E_{\text{old}}$

    if $\Delta E \leq 0$ **or** $\log \mathcal{U}(0,1) < -\Delta E$ **then**

      Accept move

    else

      $\mathbf{r}_k \leftarrow \mathbf{r}_{\text{old}}$ ▷ Reject

    end if

  end for

▷ **Phase 2: Membrane height updates**

  for each grid cell (i, j) **do**

    $h_{\text{old}} \leftarrow h_{ij}$

    $E_{\text{mol,old}} \leftarrow$ sum of E_{mol} for all molecules in cell (i, j)

    $h_{ij} \leftarrow h_{\text{old}} + \mathcal{N}(0,\, \sigma_h^2)$; **if** $h_{ij} < 0$ **then** $h_{ij} \leftarrow -h_{ij}$ ▷ Reflect to keep $h \geq 0$

    $\Delta E_{\text{bend}} \leftarrow$ local bending energy change (5-point stencil) ▷ $O(1)$ via Eq. 2

    $E_{\text{mol,new}} \leftarrow$ sum of E_{mol} for all molecules in cell (i, j) at new h_{ij}

    $\Delta E \leftarrow \Delta E_{\text{bend}} + (E_{\text{mol,new}} - E_{\text{mol,old}})$

    if $\Delta E \leq 0$ **or** $\log \mathcal{U}(0,1) < -\Delta E$ **then**

      Accept move

    else

      $h_{ij} \leftarrow h_{\text{old}}$ ▷ Reject

    end if

  end for

end for

Output: Depletion width $w = \text{median}(r^{\text{CD45}}) - \text{median}(r^{\text{TCR}})$